Detecting Hopewell Earthworks in Ohio with a Convolutional Autoencoder

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Abstract**—**This paper presents a study on unsupervised machine learning using a Convolutional Autoencoder (CAE) to detect Hopewell Culture earthworks in Ohio as anomalies in the terrain. A custom CAE was trained on 224 LiDAR Digital Elevation Model (DEM) images and evaluated on a balanced, 76-sample dataset. The base model variant achieved a weighted F1 score of 54%, a weighted precision of 73% and weighted recall of 61% with an AUC of .688. An ablation study of six model variants looked at the use of activation functions on the output of the model and application of a high-pass filter. The ablation study revealed that applying a high-pass filter to the input images inverted the anomaly detection response of the model, providing insight into the characteristics of earthwork features in LiDAR data. The base model was able to detect prominent earthworks but struggled with smaller earthworks and some terrain features. The paper concludes with recommendations for future improvements.

Keywords—unsupervised machine learning, anomaly detection, LiDAR, earthworks, CAE

GitHub Link: https://github.com/maknarfanar/earthworks-lidar

I. INTRODUCTION

Light Detection And Ranging (LiDAR) uses pulses of laser light to image land areas and structures. It has become an important tool for state and local governments to use in managing land and resources. For decades, archaeologists have used remote-sensing technology, including LiDAR to discover and survey archaeological sites. Much work has been done using this technology at sites throughout the world but the use of machine learning (ML) to review LiDAR images is a growing field [2,12]. Progress has been made with this combination for the purpose of discovering sites, surveying excavations for records, and creating 3D models of ancient structures. Few studies have examined this combination on the ancient Hopewell earthworks of Ohio.

LiDAR products are transformed through many visual techniques (VTs) to allow archaeologists to manually review images for traces of the archaeological features they are seeking and then prioritize ground-truthing [13]. VTs were designed for human perception and whether they benefit or hinder ML is an open question. This area warrants evaluation of VT use and exploration of alternatives.

A Convolutional Autoencoder (CAE) is an unsupervised machine learning model that trains on the majority, or normal, class from the data with the goal of detecting the minority class as an anomaly. In Ohio, there is a scarcity of earthwork locations and a surplus of locations without earthworks. By defining that surplus as the majority class and training a CAE on unlabeled LiDAR locations that are assumed to be in the majority class, then when presented with data that includes both majority and minority classes, the model will show a larger reconstruction error for the minority class, revealing the anomaly.

II. RELATED WORK

Guyot et al. emphasized the widespread use of VTs applied to LiDAR for archaeological interpretation [2]. The research of LiDAR on Adena-Hopewell sites in Kentucky by Henry et al. relied on a workflow where a human would manually look for anomalies after applying VT but also remarked on the challenges due to the palimpsestic nature of the region being studied [3]. Hopewell earthworks have a high variance in the ways they have been impacted by the different modes and extremes of human habitation over the almost two thousand years since they were constructed when compared to archaeological sites in Mesoamerica or Southeast Asia locations.

LiDAR technology has been successfully used with ML throughout the world. In Europe, Verschoof-van der Vaart and Lambers, applied a pretrained CNN variant for the automated detection and classification of archaeological objects in the central Netherlands [16]. In Mesoamerica, Mayan archaeology has widely used LiDAR data and ML as noted by Somrak et al., and Kokalj [7,12]. In their studies, data conditioned through VTs were used in pre-trained CNN models to classify Mayan archaeological features. In the Americas, Davis, et al., made use of publicly available LiDAR for southeastern South Carolina to apply templating methods in an automated process that narrowed down a list of over 7,000 possible sites to 15 they could focus on for ground-truth surveys. While not all of these approaches used ML, they demonstrate the growing use of automated methods in site detection.

Convolutional Autoencoders (CAEs) have been slow to enter the field of archaeology. Guyot et al. suggested the use of Deep CAEs for automatically extracting features using LiDAR [2]. Later, Jayeprokash and Gonski, and La Grassa et al. applied CAEs to satellite imagery, establishing notable techniques and results, and demonstrating CAE effectiveness for terrain-based anomaly detection [5,8].

Now a tri-fold gap exists when considering Hopewell sites in Ohio. First, LiDAR has been used for manual remote sensing in Ohio but there is scant information on the use of LiDAR with ML in this region. Second, European and Mesoamerican sites have documented ML applications. Third, CAE techniques have not been applied to the Hopewell LiDAR datasets in documented articles and reports.

III. METHODOLOGY

A. Problem Formulation

DEM tiles can be noisy with their 1000x1000 pixels covering 1,250 by 1,250 feet of real-world terrain. Despite processing the data to represent bare-earth terrain by classifying out vegetation and structures, one tile could contain a creek, a hill, a road, an agricultural field, artifacts from the LiDAR survey, and other noise elements. The size of a tile and the inherent noise from human habitation and the natural environment greatly increase the challenge of categorizing earthworks in the data [3].
Hopewell earthworks are old, with studies indicating that Hopewell earthworks are over 2,000 years old. Studies have shown that they degrade inconsistently over this span with some remaining intact and others barely evident in the landscape. Human activity, especially after European settlers came to North America, has destroyed or distorted Hopewell sites [3]. The pace of this loss creates urgency to discover more Hopewell sites.

The goal is to assist researchers in discovering potential earthwork locations that are worth ground-truthing before human progress destroys or buries them. To address this challenge, this paper presents a study on unsupervised machine learning using a Convolutional Autoencoder (CAE) to detect Hopewell Culture earthworks in Ohio as anomalies in the terrain.

B. Model Architecture

For this study, a custom autoencoder was designed to ingest 1x1,000x1,000 LiDAR DEMs. The model had the classic parts of an autoencoder: encoder, bottleneck, and decoder [5].

The encoder consisted of four pairs of Conv2d-ReLU, each with a kernel size of 3 and a stride of 2. This ingested the 1x1000x1000 DEM image and reduced it to 2x499x499, then 4x249x249, then 8x124x124, and finally to a 16x61x61 tensor.

The bottleneck was set as an additional Conv2d layer instead of a linear layer to take advantage of the group updates during the backpropagation of a convolutional layer. The bottleneck used its single Conv2d layer to compress the tensor 4x30x30 using a kernel of 3 and stride of 2.

The decoder contained five layers and the first layer after the bottleneck had the first axis jump to the 16x61x61 tensor as seen in the final layer of the encoder. The other layers in the decoder continued with the same kernel and stride as the encoder layers in reverse but with the second and fifth layers including an output padding to adjust the size of the array and finish with a tensor of 1x1,000x1,000.

The model did not use MaxPooling layers. The resolution of the target DEM files were 1.25x1.25 feet per pixel. Larger earthworks undoubtedly show up in the data as dozens or hundreds of pixels. Smaller earthworks might only be a few pixels in width or length. MaxPooling was not used out of a desire to preserve those smaller edges in the model so the CAE could have a chance to detect them as an anomaly. If included in the model, MaxPooling could wash out the smaller anomalies from the images which would be detrimental to detection.

In the base model, there is no activation function at the output of the decoder. The lack of an activation function was intended to help the model reconstruct the original image from the bottleneck encodings. However, for the ablation study a TanH and a Sigmoid were evaluated.

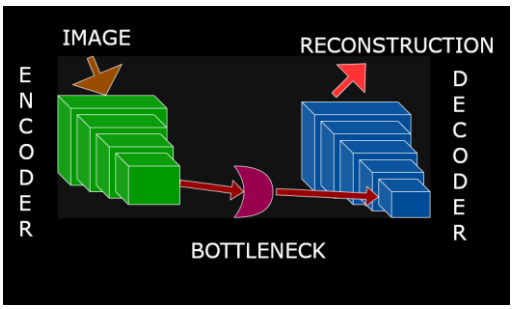

Fig. 1. Architecture overview of the proposed Convolutional Autoencoder showing the encoder, bottleneck, and decoder.

C. Training Procedure

The training pipeline used 224 DEM tiles with a .8/.2 training and validation split giving 180 and 44 tiles respectively. Both of these were loaded with a batch size of 8. When loaded, the data augmentations included horizontal flip and vertical flip transforms which worked well for this data. Other data augmentations like rotations and mirroring affected the model detrimentally during initial testing, so the two flips were chosen with a 0.5 probability for each. After these transformations, the single band for elevation was locally normalized for each tile with the z-score method. This set each value around a mean of zero with unit variance. This meant that the values of each tile were distributed around the mean of zero, but could be any reasonable value above or below the mean.

For the training loop, the Adaptive Moment Estimation (Adam) optimizer was chosen because it is a stochastic optimization method that has a proven track record when used with CNNs and CAEs [11].

To help the model train efficiently, the learning rate was set up with a scheduler to start at 1e-3 and drop to 1e-4 within the first 25 epochs. Testing prior to implementing the scheduler had shown that the model training speed could benefit from a higher starting learning rate. Weight decay was set to a default 1e-5 and testing with 1e-4 weight decay and with no decay resulted in a slightly worse model. For the loss function Mean Squared Error (MSE) was chosen as a general-purpose loss function [17]. Studies using Structural Similarity Index (SSIM) and Peak Signal-to-Noise Ratio (PSNR) will be done for a future project [8]. The limit of 100 was set on max number of epochs because according to experiments if the model did not stop training by 100 epochs then the 100 epoch limit was not significantly different from training that went beyond 100 epochs.

All code was written in Jupyter Notebook files using a Python 3.11 kernel and run in an environment with Intel(R) N100, 800 Mhz, 4 Core(s), 4 Logical processor as the CPU.

IV. EXPERIMENTAL SETUP

A. Datasets

The dataset was comprised of publicly available LiDAR DEM tiles. When LiDAR surveys are conducted the results are a point cloud of laser light returns that are then processed to classify those light returns into classes such as vegetation, structures, etc. The LiDAR tiles for this project were available as both enriched LiDAR point clouds and as DEMs. The DEMs selected for this research thus were 1000x1000 tiles with a single dimension for the elevation of the ground. Structures and vegetation had been removed.

The DEMs were available from Ohio Geographically Referenced Information Program (OGRIP) [10]. The site's main page displays an interactive map of Ohio's 88 counties and a LiDAR product selection area with up to six distinct categories of products sorted on the interactive map. The map is used to select a tile to download and then choose the type of LiDAR product. Alternatively, one could search the map by entering a city, county, or zip code or also by entering decimal latitude and longitudes.

The available products varied by county to county or region to region as part of the funding for the LiDAR products was the decision of localities on whether they would fund some of the more comprehensive surveys. As this program has existed in Ohio for many years, some locations had two or more recent years of data available for download and some had only older data. This patchwork quilt of LiDAR resources quickly self-eliminated some regions of Ohio from the study, notably most of the northern Ohio Lake Erie coastline. The DEMs for this study all come from locations that had data from 2020 to 2023. These years were most compatible with each other because they used the same or similar LiDAR equipment and shared similar tile ID styles, unlike older data available. From the OGRIP site only tiles that were under the "Elevation (3DEP+)" label were chosen. From these, the DEM tiles chosen as having the "gold standard" format were from Licking County because of the presence of the Newark Hopewell earthworks. This meant that DEMs from other locations needed to match the same meta-data tags as the Licking county DEMS: a shape of 1x1000x1000 and that a resolution of 1.25x1.25 feet per pixel equating to a tile covering 1,250 feet by 1,250 feet of land area.

Using the UNESCO World Heritage List as a reference, 38 tiles were selected based on a manual inspection of the tiles in proximity to the geographic coordinates of the earthworks [15]. Some of the tiles had numerous prominent earthworks and other tiles only had a small section with a single portion of an earthwork. Additionally, several tiles for the Van Voorhis Wall were included. The Van Voorhis Wall tiles contain fragments of earthworks that are barely detectable from a visual inspection of the DEMs. The other half of the test set were 38 tiles that were randomly selected true negative tiles. Together these tiles made for a test dataset that had 76 tiles with balanced class support on the positive (anomalous) class and the negative ("normal") class.

The training dataset was chosen with the same basic scheme as the 38 non-earthwork tiles and were selected from all over the state of Ohio except for areas in the south portion of Ohio unless they showed no obvious signs of earthworks. More than 400 tiles were downloaded for

consideration, but not all matched the requirements for resolution and number of bands. Within the tile loader class a mechanism helped screen illegitimate tiles and this reduced the eligible tiles to 224 for the training and validation datasets combined. The training dataset was split into training and validation sets with a .8/.2 split yielding 180 training tiles and 44 validation tiles to go with the 76 test dataset tiles.

TABLE I. ***Summary of Data***

Dataset	Train	Validate	Test	Classes
Main DEM Tiles	180	44	76	2

B. Evaluation Metrics

Evaluation of the results focused on the F1 score, the confusion matrix and the Area Under the Curve (AUC) score. Because CAE is an unsupervised ML technique, the model was not trained on labels, but the test set had 76 samples with 38 positive labels and 38 negative labels but needed a threshold as a decision boundary for scoring the tiles by their reconstruction errors. To avoid complications like data leakage, the validation dataset, which contained no positive labels, was used to determine a decision boundary by taking the mean of the validation set's errors and then adding two times the standard deviation. This method was used to determine the threshold for each respective trial in the ablation tests.

C. Implementation Details

The learning rate was 1e-3 with a gradual decrease over 25 epochs to 1e-4. The weight decay was 1e-5. Batch size was 8 due to the size of the dataset. Epochs were set to 100 but early stopping was implemented to stop after the validation error increased over three epochs [16].

V. RESULTS

A. Main Results

The baseline model variant started training at 1.07925 error score with a learning rate of 1e-3 that decreased with a learning schedule to 1e-4 over 25 epochs. In Figure 2 one can see the model search for its footing until epoch 7 after which it quickly reduced the error rate, demonstrating it was learning and finishing after 3 epochs of fluctuating validation error at 44 epochs.

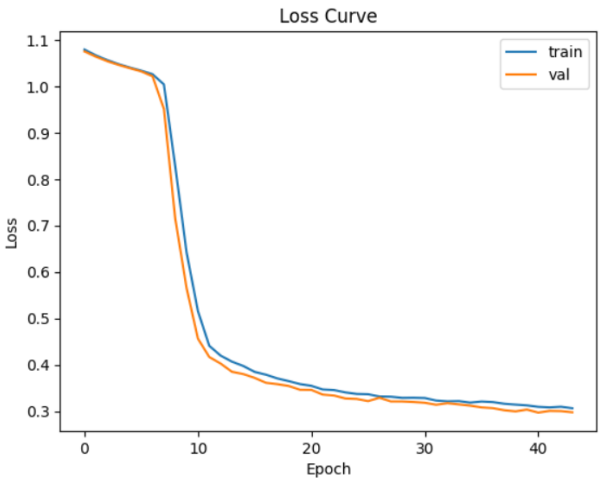

Fig. 2. Loss curve for the baseline variant

The threshold of .5240 was derived by evaluating the validation set on the trained model and then taking the mean of validation errors plus two standard deviations of the validation errors. This threshold was then applied to the results of evaluating the test set on the trained model. The confusion matrix resulting from this process is shown below in Figure 3. True Negatives were the strongest result of this model with a count of 37, but only 9 true positives. Many of the true positives fell below the threshold level. Because this is using validation data from the training set this indicates a possible issue with the training data. This topic will be explored in the qualitative analysis section of the paper.

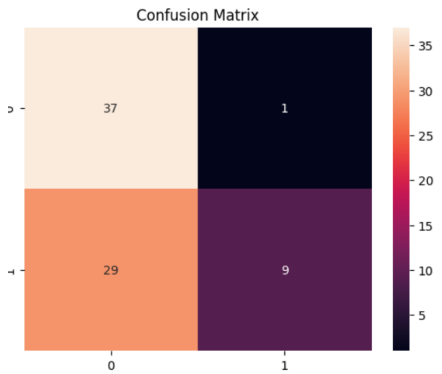

Fig. 3. Baseline Variant Confusion Matrix

This resulted in the baseline variant achieving a weighted F1 score of **54**%, with a class F1 of 71% for class 0 and 38% for class 1. The weighted precision of 73% resulted from a class precision of 56% for class 0 and 90% for class 1. Likewise, the weighted recall of 61% came from a class 0 recall of 97% and a class 1 recall of 24%. The benefits of this model are that when something is called a positive then the model at this threshold is usually correct, but only the tiles with the largest reconstruction errors are reaching this threshold.

The AUC for the baseline model variant reached .688 further showing that despite the substantial number of false negatives there is still some usefulness in the model. In general, the model has shortfalls, but the metrics inform us on what techniques could be implemented to improve the results. This will be discussed more in the qualitative analysis section.

TABLE 2. ***Summary of Ablation study with weighted scores***

Metric/ Model	AUC	F1	Precision	Recall	Epochs	Threshold
Baseline	.688	.54	.73	.61	44	.5240
TanH	.738	.56	.74	.62	29	.5537
Sigmoid	.699	.58	.75	.63	25	.814
Baseline + HPF	.216	.39	.76	.53	66	.969
TanH + HPF	.280	.39	.76	.53	100*	.956
Sigmoid + HPF	.216	.39	.76	.53	66	.969

* Model did not trigger early stopping; training ran to ablation epoch limit.

B. Ablation Study

To analyze the effect of having a decoder without an activation function and the effects of a high-pass filter (HPF), six model variants were evaluated. The first triplet is the first three variants tested with only changes in the activation function. In the first variant, the base model was as described earlier in this paper, meaning no activation function for output. In the second, a hyperbolic tangent (TanH) function was used as the final output. This meant that all outputs from the model would be between –1 and 1. The third model used a Sigmoid as the final output of the model, limiting the output to 0 to 1. Note that the images are elevation data that have been z-score normed to negative and positive values centered around a mean of zero and this does not constrain the numbers between –1 and +1 as TanH does.

The second triplet mirrored the first but with the addition of HPF applied to each test. The HPF was a transformation to each image when loading that created a blurred version of the original image using a gaussian filter which was then subtracted from the original image. This had the effect of highlighting areas of rapid change in elevation while smoothing more gradual changes [2, 7].

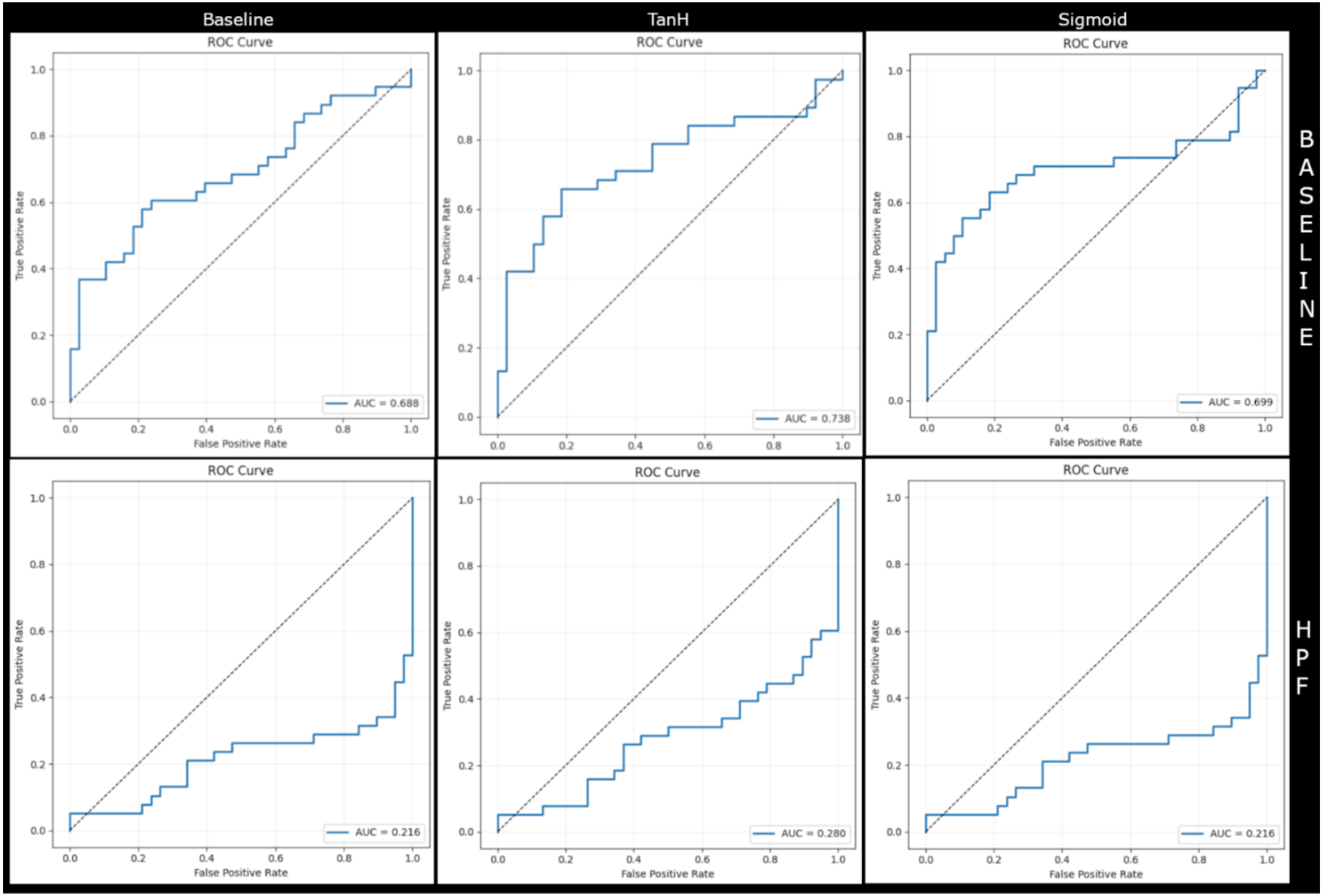

Fig. 7. ROC curve results from ablation study.

It is notable that the first triplet outperformed the second and the second triplet had an inverted AUC response to the HPF treated data as all three AUC scores were below 0.5. The second triplet's metrics were also almost identical to each other. Also, both the TanH and Sigmoid outperformed the base variant in every metric in table 2.

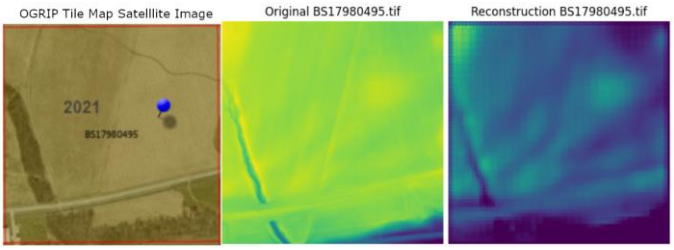

Fig. 4. The tile from the test dataset with the highest reconstruction error from baseline model variant shown in satellite image, original image with color added and reconstructed image with color added.

C. Qualitative Analysis

The tile with the greatest reconstruction error for the baseline model was tile BS17980495. This was a true positive tile that demonstrates the detection capability. Figure 4. shows a comparison between the tile's satellite image, original tile image and the reconstructed image. The color is added for contrast as the DEM is an elevation model. This is a tile from the southwest portion of the Hopewell Mounds location at 39.359553°N, 83.097184°W. One can see the creek and the road showing opposite but consistent elevations with minor change and one can make out the mounds. These features do not add to the error rate as they are visible in both the original and the reconstruction. However, there is more error shown south of the road as the reconstructed image is missing some details from the original. Cross-referencing with this tile as shown in Figure 5 shows that this area south of the road is part of the earthwork complex.

Fig. 5. Comparison of satellite image tiles for Hopewell Mounds site to the site map courtesy of Hopewell Ceremonial Earthworks [4].

Interestingly, the tile in the test dataset with the smallest reconstruction error for the baseline model is a false negative tile from the same area, BS18010496. Figure 5 shows the ground-truth layout of the site in relation to the tiles from the OGRIP site [10]. This false negative tile was below the threshold and marked as having no anomalies based on the CAE pipeline. The area of this tile has anomalous features evident from the map and satellite image in Figure 5 but examination of the original and reconstruction images as seen in Figure 6 clearly shows the low error between the original and reconstruction images, though one can see the reconstruction artifacts at the top border of the image. This tile may have benefited from a global z-score applied from the entire dataset instead of a local z-score or setting a floor that all elevations are converted to similar to the methods of template searching algorithms [14]. It also could have been processed with a standard visible technique such as hillshade which may have emphasized the visibility of elevation features for both human and computer [6].

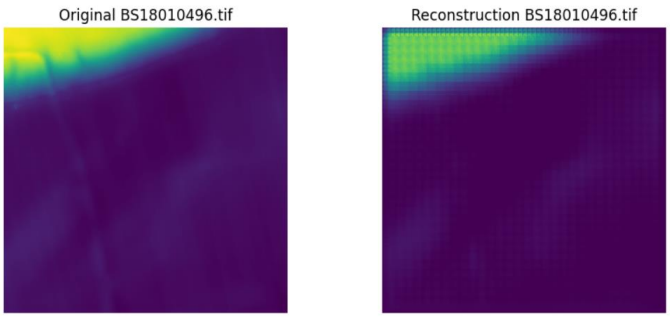

Fig. 6. Comparison of the original and reconstructed images for the test set tile with the lowest reconstruction error

Comparing the ROC curves for the ablation study reveals several notable patterns. In figure 7, the TanH had the highest AUC between the other two in the baseline and the HPF. This is likely because TanH was closest to the z-scored normed elevations, though the ranges differed. Considering the number of ReLU activations preceding it that restricted output to zero and above, this TanH function allowed the model to do less work from the convolutional layer to get the elevations to more closely match the input image's elevations.

VI. DISCUSSION

Early analysis of the DEM images before building the CAE model showed how earthworks and other elevation changes seemed to pop out of the image when high-pass filtering (HPF) transformations were applied. The HPF ablations showed the effect of including this transformation in the model. As the CAE was trained on HPF images, it learned that features that pop out after HPF treatment are normal and are not anomalies. The model learned this to such an extreme that when test data was evaluated on an HPF model, the Hopewell earthworks looked normal and were incredibly easy for the model to reconstruct. This caused the model to interpret the tiles without earthworks as the anomalies, inverting the answer-key, so to speak.

The TanH showed significant improvement in the AUC score over the base model. This was due to the z-score norm forcing elevation values around a mean of zero with both negative and positive values, which mirrors the TanH output range. A final TanH activation function allowed the model to better handle the final conversion to negative and positive values without having to try to do it with a series of ReLU functions establishing a floor of zero. That TanH allowed the model to better reconstruct the original image. Future testing of this would involve normalizing the elevations differently, perhaps with a Min-Max normalization to restrict the input to a [0,1] range and evaluate it with a sigmoid as the final output. Alternatively, the TanH function itself could be used to normalize the tiles. Additionally, the activation functions for the layers would also be converted from ReLU to TanH but this would be a notable change in the model architecture.

The model used a local z-score in which each tile was scored independently from other tiles in the dataset. It is generally accepted that global z-score is better for anomaly detection. The hypothesis for this CAE study was that the subtle earthwork signals could be more prominent with local normalization. The tile in Figure 6 calls this into question and a comparison would need to be done to determine how much it is a factor for earthworks.

There was a large variance in the amount of an earthwork contained in a tile. Earthworks like the Octagon and the Great Circle take multiple tiles to see the complete structure. Some of those tiles show the earthwork in nearly every facet of the image. Others might only have

a small fragment of the larger earthwork in a corner or border section of the tile. Smaller earthworks like the Van Voorhis Wall can also take several tiles, but only ever make up a small slice of the 1000x1000 pixel area of the tiles because they are small formations with some of their parts hard to see because of the palimpsestic nature of the terrain. Future studies could compare the size of the tile in relation to earthworks. Davis et al. used a window method in which they applied a rules-based template matching system to identify earthworks in a large area [1]. Youssef et al. also mention localized versus global reconstruction tasks as a benefit [17]. One method could be to extend this idea to identify a tile and portion of a tile by scoring grid sections of tiles for their reconstruction error with the tile reconstruction error determined from the maximum reconstruction error of the grid-squares. One could also classify tiles before scoring the grid-squares so that a tile that contains urban or suburban features could be treated differently than a forest or farmland for a multi-stage detection pipeline.

For this initial study by the author, part of the challenge was researching LiDAR and the DEM products to develop a clear data understanding of the noisy tile ecosystem and this contributed to challenges in curating an effective dataset for training. Many true negative tiles in the validation set had higher reconstruction errors than many of the tiles in the training set. This was an overlap in the two error distributions that made the validation-derived-threshold an imprecise decision boundary. A classic remedy for this is simply more training samples, and that is one option, but it would also be a larger number of error-prone tiles to train on. A more effective solution would be to develop a pipeline for validating not just the dimensions and resolution of the files but to classify the terrain to exclude tiles that are not representative of where earthworks might typically be found. This could be accomplished by using Ohio archaeological databases to simultaneously expand the number of true positives for the test set and categorize the types of areas these sites are found in.

VII. CONCLUSION

The results show that a Convolutional Autoencoder can be a useful tool in detecting Hopewell Culture earthworks in Ohio. The baseline CAE method used can detect larger earthworks with better than random effectiveness. Smaller earthworks are a challenge and terrain features can cause noise in the training set. The current model achieves a result that is better than random baseline. However, several pipeline components are limiting performance and the model can be improved upon. For example, dataset curation, model architecture, high-pass and low-pass techniques, normalization schemas, and multi-stage detection represent the likely next steps.

The use of machine learning in the study of Hopewell Earthworks is still in its infancy. While there are prominent earthworks in Ohio that attract the attention of tourists, there are smaller earthworks that remain under-documented despite their significance. Archaeologists have found evidence of a “road” dubbed the Great Hopewell Road stretching about 60 miles from the Newark, Ohio Hopewell sites to the Chillicothe, Ohio sites [9]. It is a set of parallel earthworks of which the Van Voorhis Wall is one example, and only fragments of this road have been identified along its route. This is a small part of the vast continent-scale trade network that spanned North America during the time of the Hopewell [9]. With better techniques to develop a map of candidate earthwork locations in Ohio, the study of the Great Hopewell Road and other sites could continue before palimpsestic entropy erases these sites for the rest of time.

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